

Test Assignment — AI Automation Engineer / Applied AI Engineer

General Guidelines

- You may use **any stack**: Python, JavaScript, n8n, Make, Zapier, Pipedream, LangChain, OpenAI, Claude, Vapi, Retell, Twilio, Telegram Bot API, etc.
- Please use **synthetic / test data only**
- A production deployment is **not required**. Sandbox accounts, test bots, and clearly demonstrated simulations are acceptable.
- We care more about **clarity, execution, and judgment** than polish.
- Across the whole assignment, include at least **2 real integrations**. One should be a communication or messaging integration, and one should be a data destination such as Google Sheets, Airtable, or a database.
- Since no input files will be provided, please define your own assumptions, create your own sample data, and document them clearly.
- Submission should include:
 - a 5–10 minute demo video,
 - sample outputs,
 - and a brief architecture note.

Part 1 — Parent Outreach Voice Agent

Scenario

We need to complete and clean parent/student records before scheduling group events.

Your Task

Build an AI agent that:

- call parents,
- asks for and captures missing details about their child / student, including:
 - parent email,
 - mobile phone,
 - school name,
 - preferred weekday,
 - preferred time for a group event,

- and any relevant notes,
- writes the structured result to a **Google Sheet**,
- sends a **WhatsApp follow-up** if the parent does not answer,
- records the outcome of each outreach attempt.

Assumptions

Since no files are provided, you should create your own small test dataset and define your own assumptions for:

- parent and student records,
- which fields are already known vs missing,
- available event time slots,
- outreach rules,
- and the target Google Sheet structure.

Requirements

- The agent should ask only for **missing or unclear information**, not repeat data that is already known.
- It should support these statuses:
completed / partial / callback requested / unreachable / wrong number / opt-out / human review
- Include a structured output schema with fields such as:
 - parent name,
 - student name,
 - collected fields,
 - outreach status,
 - confidence,
 - last action taken,
 - next recommended action,
 - timestamp.
- A **real Google Sheets integration** is required.
- A real voice or WhatsApp integration is preferred. A clearly demonstrated simulation is acceptable if documented well.

- If the same event is triggered twice, the workflow should **not** create duplicate rows or send duplicate follow-ups.
- Include at least one **non-LLM validation step**
- Low-confidence or ambiguous cases must go to **human review**.

Demo Cases

Please demonstrate at least 5 scenarios:

1. Parent answers and provides everything
2. Parent does not answer and gets a WhatsApp follow-up
3. Parent provides only partial information
4. Parent asks to be called later
5. Wrong number or opt-out case

Part 2 — Telegram Student Context Agent

Scenario

We want an AI agent that can automatically message students in Telegram for several use cases:

- course satisfaction survey,
- follow-up on incomplete homework,
- asking why a student missed class,
- reminding about an upcoming task or session.

The agent must understand the full context before messaging the student.

Your Task

Build an AI agent that reaches out to students on any request and automatically updates the google sheet based on the request and conversation.

Assumptions

Since no files are provided, you should define and create your own synthetic data for:

- student profiles,
- attendance history,
- homework status,

- prior interactions,
- campaign or outreach queue,
- teacher notes,
- and any lightweight knowledge base the agent may need.

Requirements

- The agent must ground its messages in the context you define, such as:
 - course name,
 - homework status,
 - attendance history,
 - prior messages,
 - teacher notes,
 - upcoming deadlines or sessions.
- It should maintain a simple human-like conversation state and avoid asking questions already answered elsewhere in the context.
- It must **not hallucinate facts** that are not present in the records or knowledge base you created.
- It should handle unclear replies and sensitive or negative replies that should be escalated to a human.
- Store the interaction history and a structured summary after each run:

Demo Cases

Please show at least 4 scenarios:

1. Homework follow-up
2. Missed-class follow-up
3. Satisfaction survey
4. Reminder about an upcoming task or session

Deliverable

- working workflow or code,
- sample Telegram conversations or sandbox bot demo,
- stored outputs / summaries,

- short demo,
 - architecture note.
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Part 3 — Reliability, QA, and Ops Layer

Suggested time: 3 hours

Your Task

Build a lightweight reliability layer on top of one or both agents.

Requirements

Define your own edge cases and failure scenarios, then implement at least **4** of the following:

- retry logic / backoff,
- idempotency,
- confidence thresholds,
- human review queue,
- quiet hours / do-not-disturb logic,
- opt-out suppression,
- audit logs,
- error logging and alert-ready fields,
- fallback behavior when a channel fails.

Also include:

- at least **10 test cases** with expected outcomes,
- a short note on how you would reduce **cost and latency**,
- one short **incident note / postmortem** for a hypothetical failure.

Deliverable

- updated workflow or shared reliability module,
- test matrix,
- short reliability note.